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(54) **METHOD FOR DISPLAYING
STEREOSCOPIC SCENE AND ELECTRONIC
DEVICE**

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See application file for complete search history.

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(56) **References Cited**

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U.S. PATENT DOCUMENTS

5,459,846 A * 10/1995 Hyatt G07G 1/12
711/E12.004
12,201,485 B2 * 1/2025 McDowall A61B 90/37
(Continued)

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FOREIGN PATENT DOCUMENTS

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CN 113434049 A 9/2021
TW 201315243 A * 4/2013 H04N 13/161

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OTHER PUBLICATIONS

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(57) **ABSTRACT**

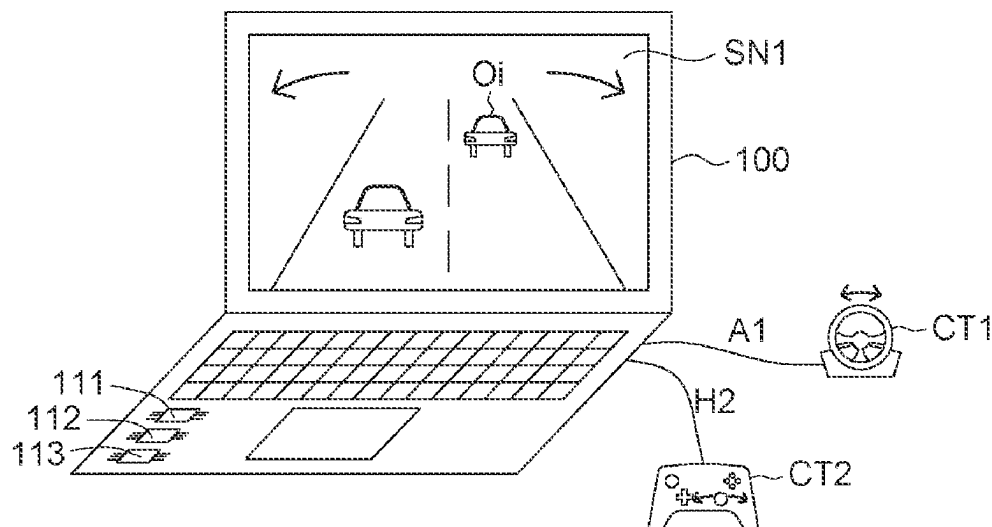
(51) **Int. Cl.**
G06F 3/038 (2013.01)
A63F 13/23 (2014.01)
G06F 3/023 (2006.01)
H04N 13/398 (2018.01)
(Continued)

A method for displaying a stereoscopic scene and an elec-
tronic device are provided. The electronic device includes a
displaying unit, an inputting unit, a stereo scene generating
unit, a controlling unit and an imaging unit. The inputting
unit includes a hot key. If the hot key is pressed, the
stereoscopic scene generating unit generates the stereo-
scopic scene comprising a plurality of frames. The control-
ling unit is used for receiving a human interface device
(HID) signal from the inputting unit, and converting the HID
signal into an angle signal. The imaging unit is used for
displaying one of the frames of the stereoscopic scene on the
displaying unit according to the angle signal.

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G06F 3/0338 (2013.01)

(58) **Field of Classification Search**
CPC . G06F 3/038; G06F 3/02; G06F 3/023; G06F
3/0338; A63F 13/23; A63F 13/803; H04N

12 Claims, 7 Drawing Sheets



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(56) **References Cited**

U.S. PATENT DOCUMENTS

2007/0019936 A1 * 1/2007 Birkenbach G03B 41/00
396/14
2013/0044191 A1 * 2/2013 Matsumoto H04N 13/117
348/E13.074
2013/0076736 A1 * 3/2013 Seol H04N 13/128
345/419
2020/0294209 A1 * 9/2020 Pohl G06T 5/50
2021/0076529 A1 3/2021 Pachoud et al.
2022/0122328 A1 4/2022 Giovannetti et al.

* cited by examiner

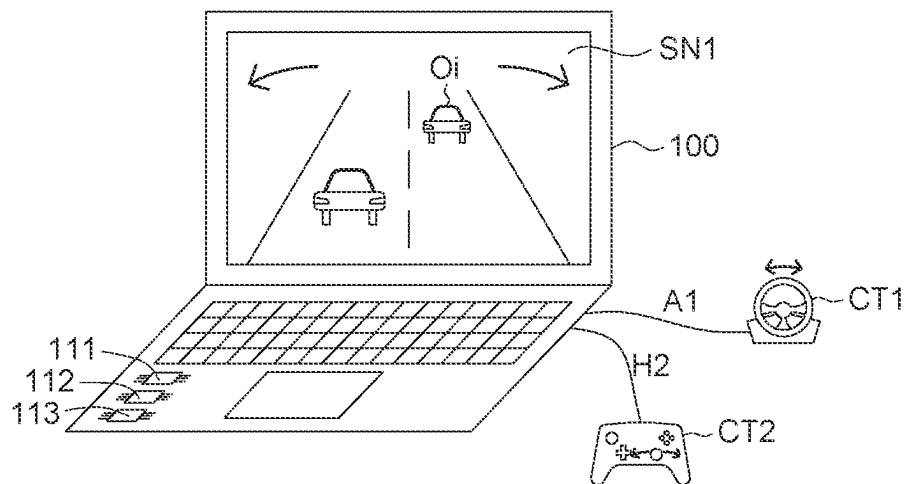


FIG. 1A

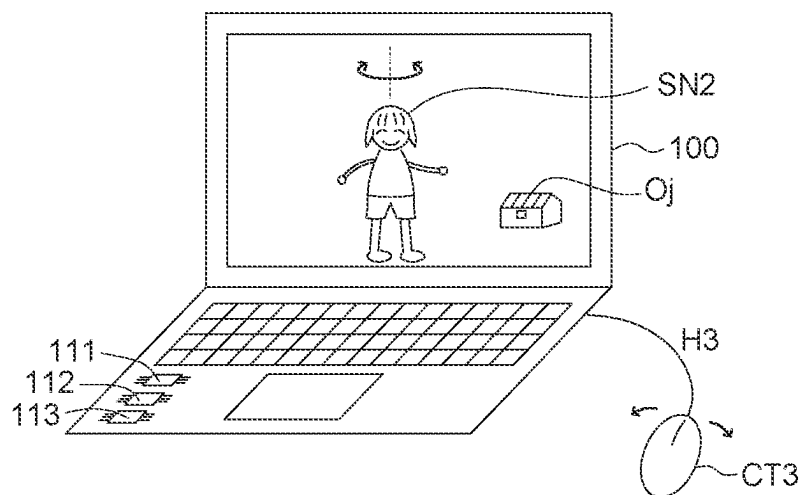
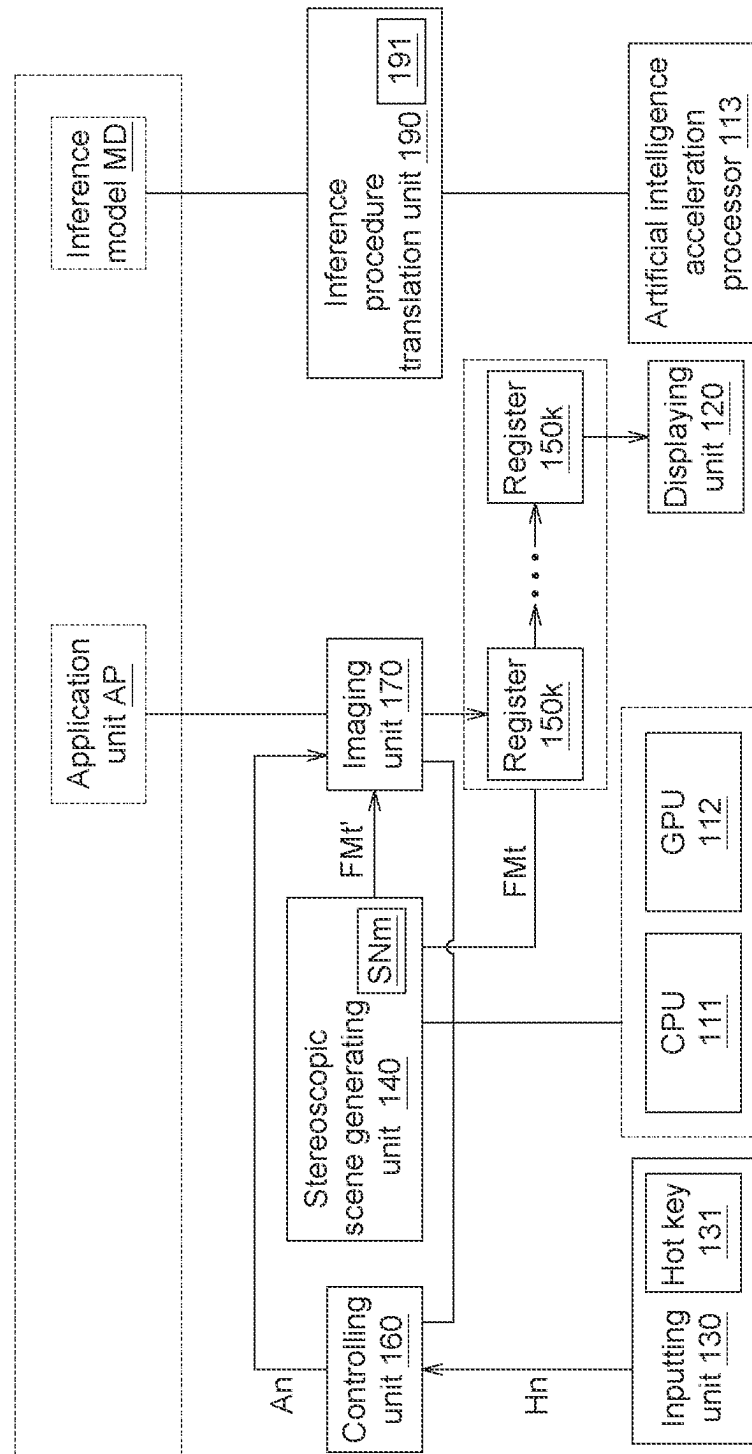


FIG. 1B

100



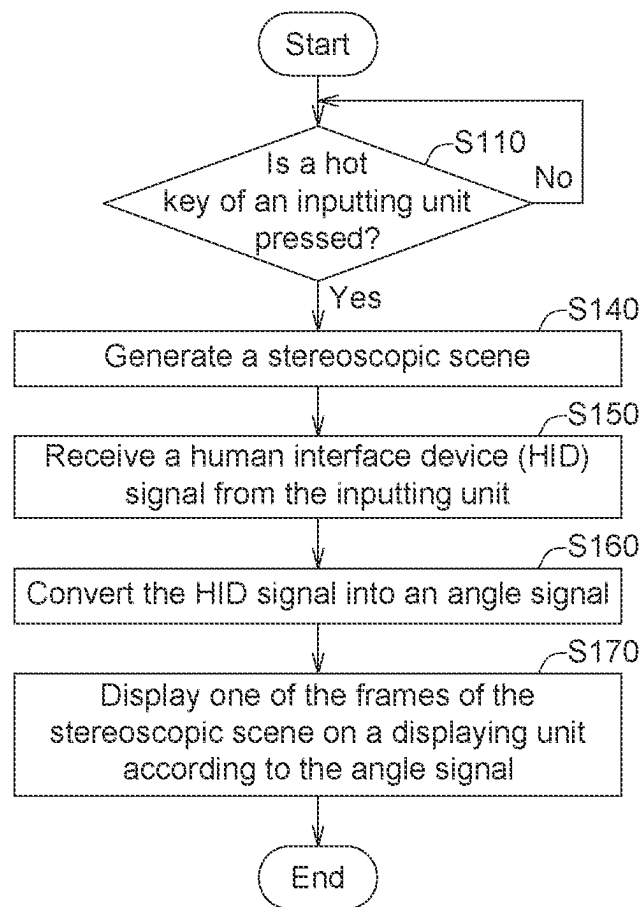


FIG. 3

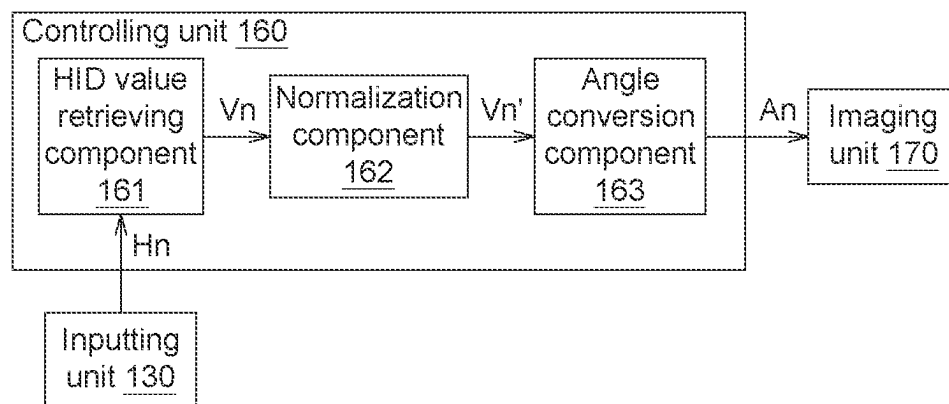


FIG. 4

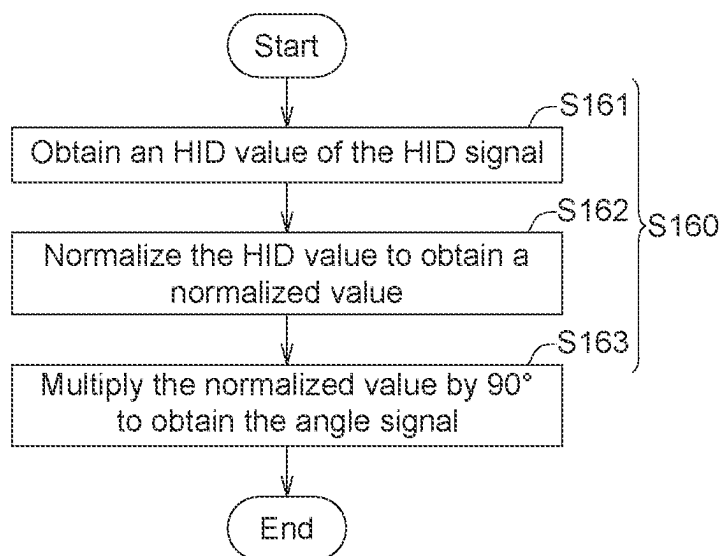


FIG. 5

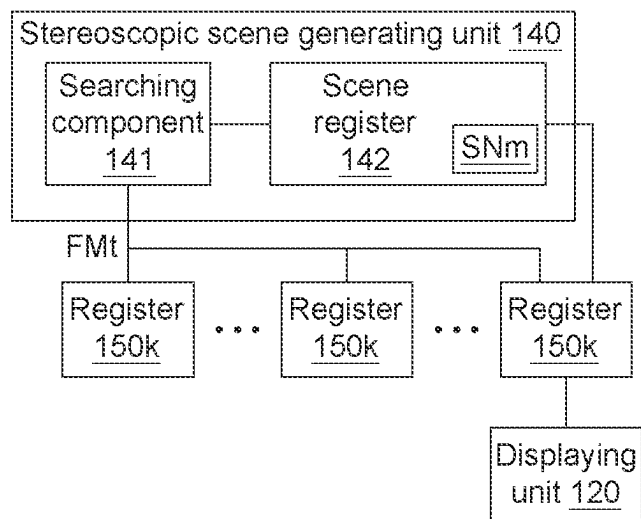


FIG. 6

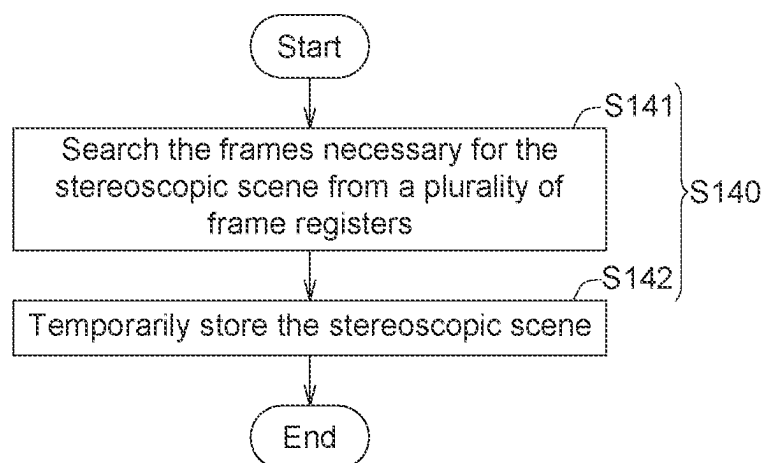


FIG. 7

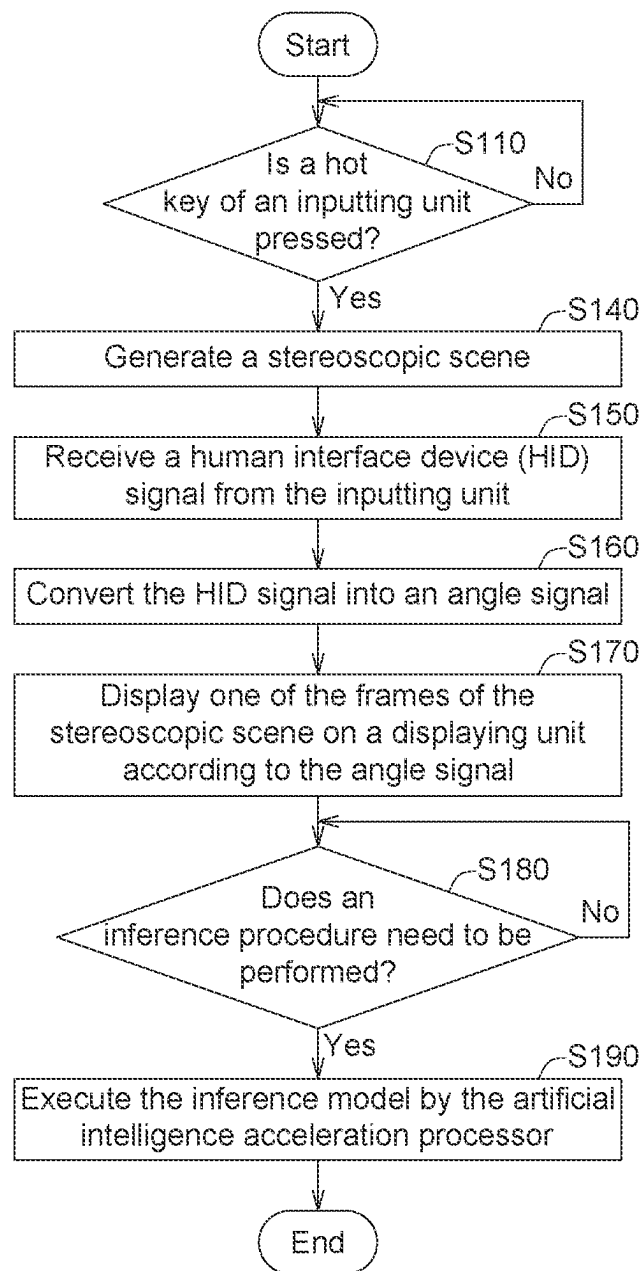


FIG. 8

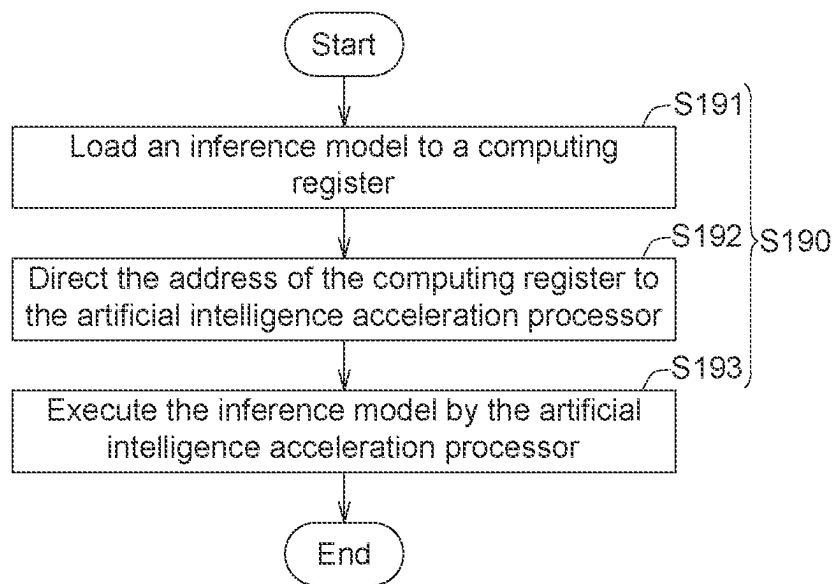


FIG. 9

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METHOD FOR DISPLAYING STEREOSCOPIC SCENE AND ELECTRONIC DEVICE

This application claims the benefit of Taiwan application
Serial No. 112123379, filed Jun. 21, 2023, the subject matter
of which is incorporated herein by reference.

BACKGROUND OF THE INVENTION

Field of the Invention

The invention relates in general to a displaying method
and an electronic device, and more particularly to a method
for displaying a stereoscopic scene and an electronic device.

Description of the Related Art

Along with the rapid development in the multimedia and
gaming industry, the user can operate stereoscopic scene
with a specifically matched remote device. For instance, the
user can view different angles of virtual frames through the
use of a head mounted display. However, without the
specifically matched remote device, the user will be unable
to operate the stereoscopic scenes.

SUMMARY OF THE INVENTION

The invention is directed to a method for displaying a
stereoscopic scene and an electronic device. Through the use
of HID angle conversion technology, the HID signal of the
inputting unit can be converted into an angle signal, so that
the stereoscopic scene can be operated with various input-
ting units. Besides, through the use of a computational
balancing mechanism, the AI inference procedure can be
oriented to the artificial intelligence acceleration processor,
so that the loading of the CPU or the GPU can be reduced.

According to a first aspect of the present invention, a
method for displaying a stereoscopic scene is provided. The
displaying method includes the following steps. Whether a
hot key of an inputting unit is pressed is determined: if the
hot key is pressed, the stereoscopic scene comprising a
plurality of frames is generated. A human interface device
(HID) signal is received from the inputting unit. The HID
signal is converted into an angle signal. One of the frames
of the stereoscopic scene is displayed on the displaying unit
according to the angle signal.

According to a second aspect of the present invention, an
electronic device is provided. The electronic device includes
a displaying unit, an inputting unit, a stereo scene generating
unit, a controlling unit and an imaging unit. The inputting
unit includes a hot key. If the hot key is pressed, the
stereoscopic scene generating unit generates a stereoscopic
scene comprising a plurality of frames. The controlling unit
is used for receiving a human interface device (HID) signal
from the inputting unit and converting the HID signal into an
angle signal. The imaging unit is used for displaying one of
the frames of the stereoscopic scene on the displaying unit
according to the angle signal.

The above and other aspects of the invention will become
better understood with regard to the following detailed
description of the preferred but non-limiting embodiment(s).
The following description is made with reference to the
accompanying drawings.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1A is an example of operating a particular stereo-
scopic scene with an electronic device.

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FIG. 1B is an example of operating a particular stereo-
scopic scene with an electronic device.

FIG. 2 is a block diagram of an electronic device accord-
ing to an embodiment.

FIG. 3 is a flowchart of a method for displaying a
stereoscopic scene according to an embodiment.

FIG. 4 is a detailed block diagram of a controlling unit
according to an embodiment.

FIG. 5 is a detailed flowchart of step S160 according to an
embodiment.

FIG. 6 is a detailed block diagram of a stereoscopic scene
generating unit according to an embodiment.

FIG. 7 is a detailed flowchart of step S140 according to an
embodiment.

FIG. 8 is a flowchart of a method for displaying a
stereoscopic scene according to another embodiment.

FIG. 9 is a detailed flowchart of step S190 according to an
embodiment.

DETAILED DESCRIPTION OF THE INVENTION

Referring to FIG. 1A, an example of operating a particular
stereoscopic scene SN1 with an electronic device 100 is
shown. The stereoscopic scene SN1 can be realized by a
space surround image. The stereoscopic scene SN1 can be
rotated either manually or automatically to display different
directions of an image. When the stereoscopic scene SN1
rotates, the object Oi moves accordingly. As indicated in
FIG. 1A, the stereoscopic scene SN1 can be realized by the
scene of a racing game, and by default setting, the stereo-
scopic scene SN1 is operated with a game steering wheel
CT1. When the game steering wheel CT1 rotates, an angle
operating signal A1 can be inputted to rotate the stereoscopic
scene SN1. Ordinary peripheral inputting units (such as
game joystick CT2) only allow the input of human interface
device (HID) signal H2, which does not match the racing
game, therefore cannot operate the stereoscopic scene SN1.

In the present embodiment, through the use of HID angle
conversion technology, the HID signal H2 of whatever
peripheral inputting units (such as game joystick CT2) can
be converted into an angle signal for operating the stereo-
scopic scene SN1.

Referring to FIG. 1B, an example of operating a particular
stereoscopic scene SN2 with an electronic device 100 is
shown. The stereoscopic scene SN2 can be realized by a
three-dimensional figure. The stereoscopic scene SN2 can
be rotated either manually or automatically to display direc-
tions of an image. In the present embodiment, through the
use of HID angle conversion technology, the HID signal H3
of whatever peripheral inputting units (such as game joy-
stick CT3) can be converted into an angle signal for oper-
ating the stereoscopic scene SN2.

Besides, during the process of displaying and operating
the stereoscopic scenes SN1 and SN2, the central processing
unit (CPU) 111 or the graphics processing unit (GPU) 112
experiences a heavy loading. Meanwhile, if the object Oj
additionally requires recognition inference, the CPU 111 or
the GPU 112 will be overloaded. In the present embodiment,
through the use of a computational balancing mechanism,
the AI inference procedure is oriented to the artificial
intelligence acceleration processor 113, so that the loading
of the CPU 111 or the GPU 112 can be reduced.

Referring to FIG. 2, a block diagram of an electronic
device 100 according to an embodiment is shown. The
electronic device 100 includes the said CPU 111, the said
GPU 112, the said artificial intelligence acceleration proces-

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sor **113**, a displaying unit **120**, an inputting unit **130**, a stereoscopic scene generating unit **140**, a plurality of registers **150k**, a controlling unit **160**, an imaging unit **170** and an inference procedure translation unit **190**. The artificial intelligence acceleration processor **113** can be realized by a neural-network processing unit (NPU) or an intelligence processing unit (IPU). The artificial intelligence acceleration processor **113** is different from the GPU **112**. The artificial intelligence acceleration processor **113** is equipped with an in memory computing function or an in memory searching structure and is applicable to the AI inference procedure.

The displaying unit **120** can be realized by a liquid crystal display panel, an OLED panel, an e-paper, a head mounted display, an AR glasses or a projector. The inputting unit **130** can be realized by a keyboard, a mouse or a joystick. The inputting unit **130** is not limited to a single device; the inputting unit **130** can also be formed of a plurality of devices. The stereoscopic scene generating unit **140**, the controlling unit **160**, the imaging unit **170**, and the inference procedure translation unit **190** are used for executing various procedures, such as processing procedure, computing procedure, analytic procedure, and judgment procedure, and can be realized by a circuit, a circuit board, a chip, or a storage device for storing program code. The register **180k** is used for temporarily storing data, and can be realized by a memory or a storage block of a hard disc. In the present embodiment, through the use of HID angle conversion technology, the controlling unit **160** can convert an HID signal H_n of the inputting unit **130** into an angle signal A_n for operating the stereoscopic scene SN_m . Also, in the present embodiment, through the use of a computational balancing mechanism, the inference procedure translation unit **190** can orient the AI inference procedure to the artificial intelligence acceleration processor **113**, so that the loading of the CPU **111** or the GPU **112** can be reduced. The operations of each element are disclosed below with accompanying drawings.

Referring to FIG. 3, a flowchart of a method for displaying a stereoscopic scene SN_m according to an embodiment is shown. In step S110, a hot key **131** of the inputting unit **130** is pressed is determined by the controlling unit **160**. The hot key **131** is a pre-defined key or a pre-defined combination of a plurality of keys. When the user requires the stereoscopic scene SN_m , the user can press the hot key **131**. In an embodiment, the hot key **131** can be realized by a particular function key disposed on the keyboard. If the hot key **131** is pressed, the method proceeds to step S140.

In step S140, the stereoscopic scene SN_m is generated by the stereoscopic scene generating unit **140**. The stereoscopic scene SN_m consists of a plurality of frames FM_t . The stereoscopic scene SN_m can be realized by a particular space surround image (such as the stereoscopic scene SN_1 of FIG. 1); the stereoscopic scene SN_m can be a particular three-dimensional figure or object (such as the stereoscopic scene SN_2 of FIG. 2).

Then, the method proceeds to step S150, an HID signal H_n is received from the inputting unit **130** by the controlling unit **160**. The device which transmits the HID signal H_n can be different from the device equipped with the hot key **131**. For instance, the device which transmits the HID signal H_n can be an external joystick, a mouse, an external touch panel, a game handle or a similar device; the device equipped with the hot key **131** can be realized by the keyboard or in-built touch panel of the device.

Then, the method proceeds to step S160, the HID signal H_n is converted into the angle signal A_n by the controlling unit **160**.

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Refer to FIG. 4 and FIG. 5. FIG. 4 is a detailed block diagram of a controlling unit according to an embodiment. FIG. 5 is a detailed flowchart of step S160 according to an embodiment. The controlling unit **160** includes an HID value retrieving component **161**, a normalization component **162** and an angle conversion component **163**. Step S160 includes steps S161 to S163. In step S161, an HID value V_n of the HID signal H_n is obtained by the HID value retrieving component **161** of the controlling unit **160**. For instance, when the HID signal H_n is received from a press key, the HID value V_n can be 0 or 1; when the HID signal H_n is received from a joystick mushroom head, the HID value V_n can be decimal values in the X direction and the Y direction; when the HID signal H_n is received from the touch panel, the HID value V_n can be the coordinate value of the X axis and Y axis.

Then, the method proceeds to step S162, the HID value V_n is normalized by the normalization component **162** of the controlling unit **160** to obtain a normalized value V_n' . For instance, the normalization component **162** normalizes the HID value V_n as a normalized value V_n' between -1 to 1 . For instance, the maximum possible value of HID value V_n is normalized as 1 ; the minimum possible value of HID value V_n is normalized as -1 . Or, the normalization component **162** can be normalized as two normalized values V_n' in the X direction and the Y direction according to the X axis and the Y axis, respectively.

Then, the method proceeds to step S163, the normalized value V_n' is multiplied by 90° by the angle conversion component **163** of the controlling unit **160** to obtain the angle signal A_n . That is, the normalized value V_n' being "1" will be converted to the angle signal A_n whose value is " 90° "; the normalized value V_n' whose value is " -1 ", will be converted to the angle signal A_n whose value is " -90° "; the normalized value V_n' whose value is " 0.5 " will be converted to the angle signal A_n whose value is " 45° ", and the rest can be obtained by the same analogy.

Through steps S161 to S163, the controlling unit **160** can convert the HID signal H_n into the angle signal A_n .

Next, the method proceeds to step S170 of FIG. 3, a particular frame FM_t' of the stereoscopic scene SN_m is displayed on the displaying unit **120** by the imaging unit **170** according to the angle signal A_n . During the operation process of the user, different HID signals H_n generated by the inputting unit **130** can correspond to different angle signals A_n ; the frame FM_t' of the stereoscopic scene SN_m that corresponds to a particular angle can be displayed on the displaying unit **120**. Thus, through steps S110 to S170, the user can display the stereoscopic scene SN_m with any inputting unit **130**.

Refer to FIG. 6 and FIG. 7. FIG. 6 is a detailed block diagram of a stereoscopic scene generating unit according to an embodiment. FIG. 7 is a detailed flowchart of step S140 according to an embodiment. The stereoscopic scene generating unit **140** includes a searching component **141** and a scene register **142**. In an embodiment, when the imaging unit **170** performs imaging, graphics information can be received from an external server connected to an application unit AP. The electronic device **100** is without the stereoscopic scene SN_m , therefore requires the stereoscopic scene generating unit **140** for generating the stereoscopic scene SN_m .

In step S141, the searching component **141** of the stereoscopic scene generating unit **140** searches the frame FM_t necessary for the stereoscopic scene SN_m from the frame register **150k**. During the operation of the application unit AP, the frames need to be decoded and rendered, and

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different angles of the frame FMt are temporarily stored in the frame register **150k**, so that the searching component **141** can collect different angles of frame FMt from the registers **150k**.

Then, the method proceeds to step **S142**, the frames FMt are temporarily stored in the scene register **142** of the stereoscopic scene generating unit **140** to form the stereoscopic scene SNm.

Through steps **S141** to **S142**, the stereoscopic scene generating unit **140** can generate the stereoscopic scene SNm.

Referring to FIG. **8**, a flowchart of a method for displaying a stereoscopic scene SNm according to another embodiment is shown. The displaying method illustrated in FIG. **8** further includes steps **S180** to **S190**. In steps **S110** to **S170**, the process of displaying and operating the stereoscopic scene SNm needs to be computed with the CPU **111** or the GPU **112**. Meanwhile, the execution of the AI inference procedure will make the CPU **111** or the GPU **112** overloaded. In the present embodiment, through the use of a computational balancing mechanism of steps **S180** to **S190**, the AI inference procedure is oriented to the artificial intelligence acceleration processor **113**, so that the loading of the CPU **111** or the GPU **112** can be reduced.

In step **S180**, whether an inference procedure needs to be performed is determined by the inference procedure translation unit **190**. If the inference procedure needs to be performed, the method proceeds to step **S190**.

In step **S190**, the computational balancing mechanism is activated by the inference procedure translation unit **190** to control the electronic device **100** to perform the inference procedure by the artificial intelligence acceleration processor **113**.

Referring to FIG. **9**, a detailed flowchart of step **S190** according to an embodiment is shown. step **S190** includes steps **S191** to **S193**. In step **S191**, an inference model MD is loaded to a computing register **191** by the inference procedure translation unit **190**.

Next, In the method proceeds to **S192**, the address of the computing register **191** is directed to the artificial intelligence acceleration processor **113** by the inference procedure translation unit **190**.

Then, the method proceeds to step **S193**, the inference model MD is executed by the artificial intelligence acceleration processor **193** to perform the inference procedure. Through steps **S191** to **S193**, during the process of displaying and operating the stereoscopic scene SNm, the AI inference procedure will automatically be oriented to the artificial intelligence acceleration processor **193**, so that the loading of the CPU **111** or the GPU **112** will not be increased.

According to the above embodiments, through the use of HID angle conversion technology, the HID signal Hn of the inputting unit **130** can be converted into an angle signal An for operating the stereoscopic scene SNm. Besides, according to the above embodiments, through the use of a computational balancing mechanism, the AI inference procedure can be oriented to the artificial intelligence acceleration processor **113**, so that the loading of the CPU **111** or the GPU **112** can be reduced.

While the invention has been described by way of example and in terms of the preferred embodiment(s), it is to be understood that the invention is not limited thereto. On the contrary, it is intended to cover various modifications and similar arrangements and procedures, and the scope of the appended claims therefore should be accorded the broad-

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est interpretation so as to encompass all such modifications and similar arrangements and procedures.

What is claimed is:

1. A method for displaying a stereoscopic scene, comprising:

determining whether a hot key of an inputting unit is pressed;
if the hot key is pressed, generating the stereoscopic scene comprising a plurality of frames;
receiving a human interface device (HID) signal from the inputting unit;
converting the HID signal into an angle signal; and
displaying one of the frames of the stereoscopic scene on a displaying unit according to the angle signal;
wherein the step of converting the HID signal into the angle signal comprises:
obtaining an HID value of the HID signal;
normalizing the HID value to obtain a normalized value; and
multiplying the normalized value by 90° to obtain the angle signal.

2. The method for displaying the stereoscopic scene according to claim 1, wherein the normalized value is between -1 to 1.

3. The method for displaying the stereoscopic scene according to claim 1, wherein the step of generating the stereoscopic scene comprises:

searching the frames for the stereoscopic scene from a plurality of frame registers; and
storing the stereoscopic scene temporarily.

4. The method for displaying the stereoscopic scene according to claim 1, further comprising:

when the stereoscopic scene is being generated and an inference procedure needs to be performed, performing the inference procedure by an artificial intelligence acceleration processor.

5. The method for displaying the stereoscopic scene according to claim 4, wherein the artificial intelligence acceleration processor is a neural-network processing unit (NPU) or an intelligence processing unit (IPU).

6. The method for displaying the stereoscopic scene according to claim 4, comprising:

loading an inference model to a computing register;
directing an address of the computing register to the artificial intelligence acceleration processor; and
executing the inference model by the artificial intelligence acceleration processor to perform the inference procedure.

7. An electronic device, comprising:

a displaying unit;
an inputting unit, comprising a hot key;
a stereoscopic scene generating unit, wherein if the hot key is pressed, the stereoscopic scene generating unit generates a stereoscopic scene comprising a plurality of frames;
a controlling unit, used for receiving a human interface device (HID) signal from the inputting unit and converting the HID signal into an angle signal; and
an imaging unit, used for displaying one of the frames of the stereoscopic scene on the displaying unit according to the angle signal;

wherein the controlling unit comprises:

an HID value retrieving component, used for obtaining an HID value of the HID signal;
a normalization component, used for normalizing the HID value to obtain a normalized value; and

an angle conversion component, used for multiplying the normalized value by 90° to obtain the angle signal.

8. The electronic device according to claim 7, wherein the normalized value is between -1 to 1. 5

9. The electronic device according to claim 7, wherein the stereoscopic scene generating unit comprises:

a searching component, used for searching the frames for the stereoscopic scene from a plurality of frame registers; and 10

a scene register, used for temporarily storing the stereoscopic scenes.

10. The electronic device according to claim 7, further comprising:

an artificial intelligence acceleration processor; and 15

an inference procedure translation unit, wherein when the stereoscopic scene is being generated and an inference procedure needs to be performed, the inference procedure translation unit performs the inference procedure by the artificial intelligence acceleration processor. 20

11. The electronic device according to claim 10, wherein the artificial intelligence acceleration processor is a neural-network processing unit (NPU) or an intelligence processing unit (IPU).

12. The electronic device according to claim 10, wherein 25
the inference procedure translation unit loads an inference model to a computing register and directs an address of the computing register to the artificial intelligence acceleration processor, so that the artificial intelligence acceleration processor executes the inference model to perform the 30
inference procedure.

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